

5. Review of 1225

5.2. The Definite Integral

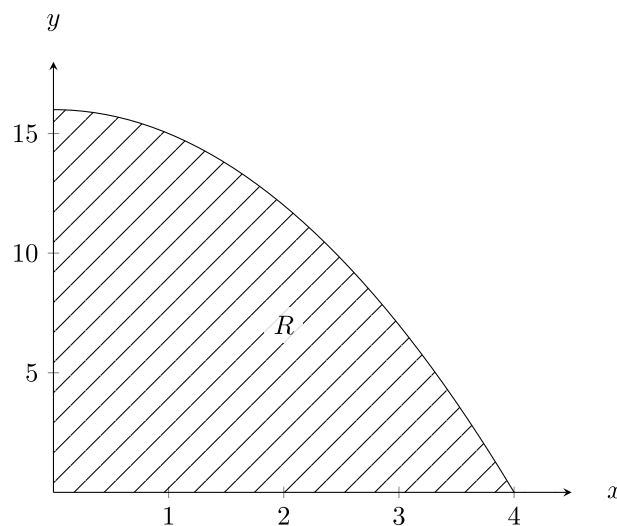
In this section, we will primarily review Riemann sums. This is because Riemann sums are used to derive many of the techniques used in the upcoming sections.

5.2 Learning Objectives:

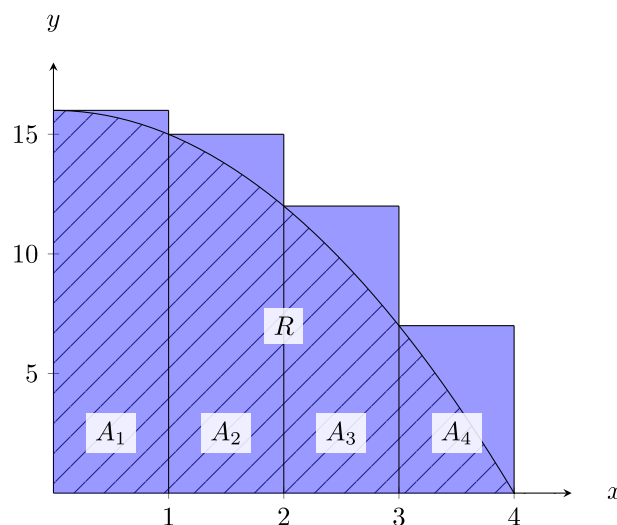
- I can approximate the (signed/unsigned) area under a curve using different types of Reimann sums.
- I can use more rectangles/sub-intervals to get a more accurate approximation.
- I know the definition of a definite integral in terms of the limit or Reimann sums.
- I can rewrite a limit of Reimann sums as a definite integral, and back.

5.2.1. Motivation

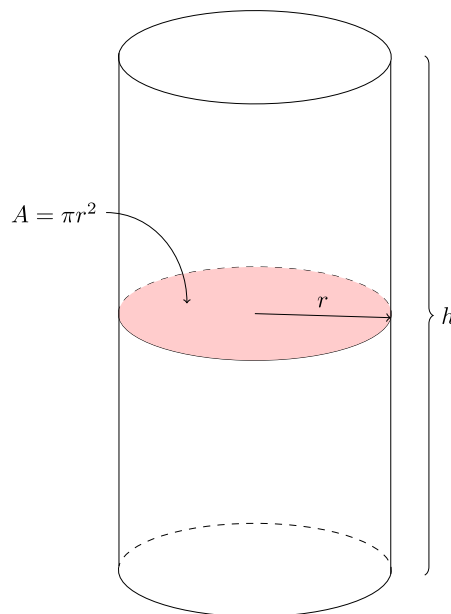
Suppose we wanted to calculate the area of a shaded region R that lies above the x -axis, below the graph of $f(x) = -x^2 + 16$, and between the vertical lines $x = 0$ and $x = 4$.



How might we estimate this area with rectangles?



The area can be estimated by summing the areas of the rectangles $R \approx A_1 + A_2 + A_3 + A_4$.

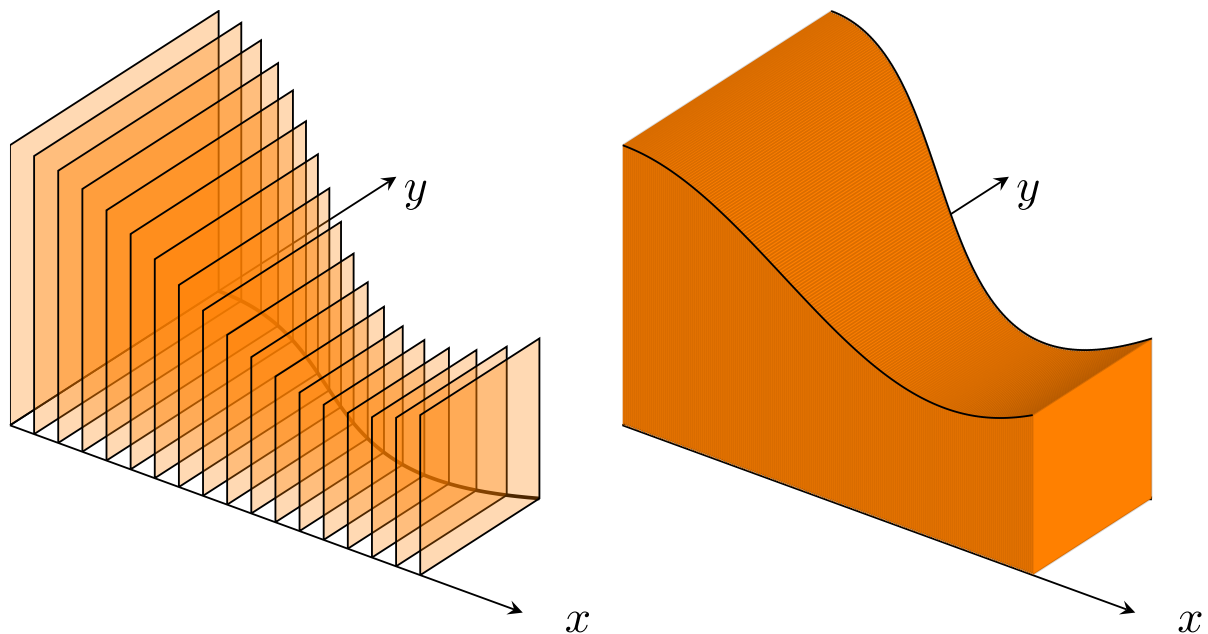


6.2.1. Slicing by Parallel Planes

Definition 6.2.1

We define a *cross-section* of a solid S as the plane region formed by intersecting S with a plane. ♣

We can build a (3 dimensional) solid S by taking the same type of cross-section (e.g. squares) for each value of x , based on some region in the Cartesian plane.

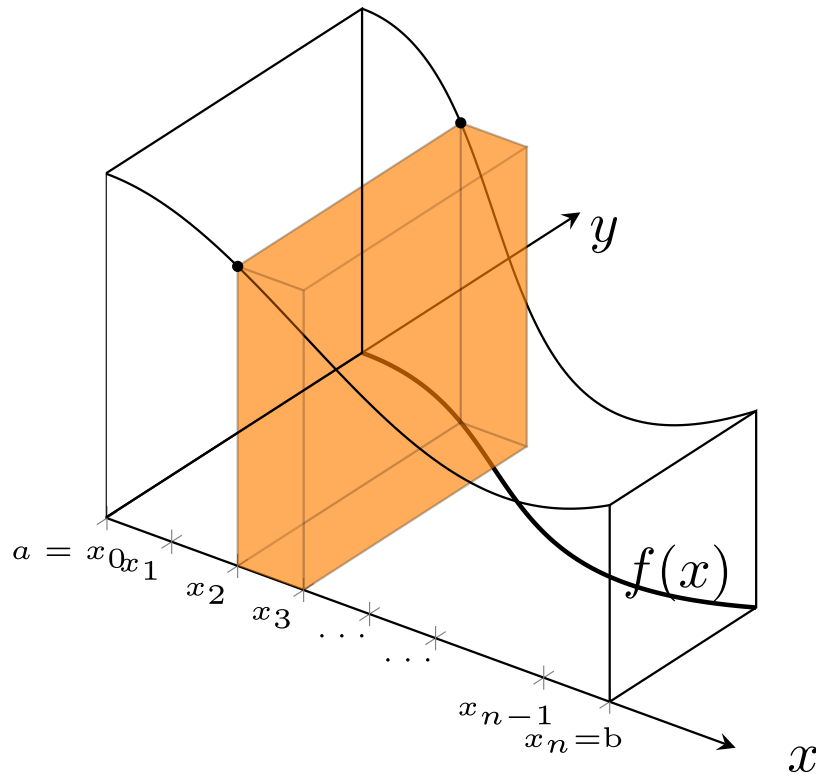


Suppose that we partition $[a, b]$ into subintervals

$$a = x_0 < x_1 < x_2 < x_3 < \dots < x_n = b$$

where each subinterval has width Δx . What is the volume of one slice of our solid? How would we use this to calculate the volume of the entire solid?

To approximate the volume, we can think of a cross-section with width Δx , then its volume is $V = (\text{Area of section})\Delta x$. Here the area of the cross-section is the area of a square with side length equal to the function value, $y = f(x)$.



As the width of the sub-intervals goes to zero and we add more rectangular prisms together, we get a better approximation of the volume. Once again, the limit of this sum ends up being a definite integral.

Definition 6.2.2

If the cross-section of a solid S at each point x in $[a, b]$ is a region $S(x)$ of area $A(x)$, where $A(x)$ is a continuous function of x , then

$$V = \int_a^b A(x) \, dx.$$

💡 Tip

To calculate the volume of a solid

1. Sketch a picture of the region and a typical cross-section.
 - Don't overcomplicate the cross-section, it should just be a prism with straight sides and top/bottom (or left/right sides) the same shape with area $A(x)$ (the area changes for each cross-section but is the same on the 2 faces of the single cross-section). It is never a part of a cone or pyramid or sphere,,
2. Find a formula for $A(x)$.
3. Find the limits of integration.

- Think about stretching the face with area $A(x)$ from one end of the shape to the other, where do you start/stop this process?
4. Integrate to find the volume, mind your units if applicable.

Note

Typical cross-sectional area formulae:

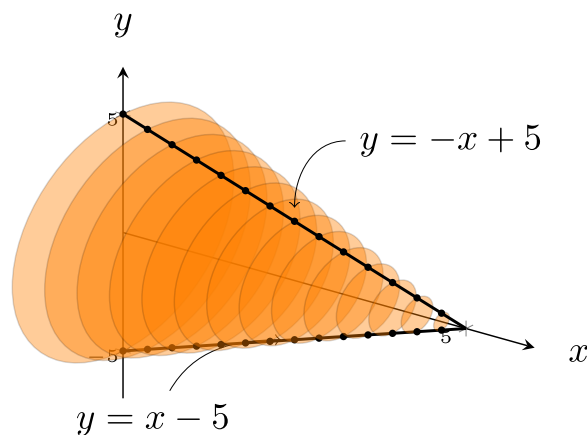
- Square: for a square with side length s , the area is $A = s^2$.
- Triangle: for a triangle with base b and height h , the area is $A = \frac{1}{2}bh$.
- Circle: for a circle with radius r , the area is $A = \pi r^2$.
- Semicircle: for a circle with radius r , the area is $A = \frac{1}{2}\pi r^2$.

Example 6.2.3

A cone with height 5 cm has a circular base with radius 5 cm. Set up a definite integral and evaluate it to find the volume of the cone.

Solution

A sketch of the volume, showing cross-sections, might look like:



We can think of slices of the cone being circles, with radius equal to the y value in $y = -x + 5$. This comes from the edge of the cone making the line $y = -x + 5$ in the x - y plane.

Then the area of one slice is $A(x) = \pi(-x + 5)^2$. The slices range from $x = 0$ to $x = 5$.

The definite integral we need to evaluate is

$$\begin{aligned}
 V &= \int_0^5 \pi(-x+5)^2 dx \\
 &= \pi \int_0^5 x^2 - 10x + 25 dx \\
 &= \pi \left[\frac{1}{3}x^3 - 5x^2 + 25x \right]_0^5 \\
 &= \pi \left[\frac{1}{3}(5)^3 - 5(5)^2 + 25(5) \right] \\
 &= \frac{125}{3}\pi
 \end{aligned}$$

We are told the distances are measured in centimeters, so the final answer is the volume is $\frac{125\pi}{3}$ centimeters cubed.

There are 2 checks we can do at the end of the problem. First, our solution is positive, which we expect because volume is positive. Second, this is a cone and we do know the volume formula for a cone:

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(5)^2(5) = \frac{125}{3}\pi.$$

This is the same thing we got from integrating, so our answer is correct!

Exercise 6.2.4

Derive the formula for the volume of a cone using similar techniques as above. *Hint: the line connecting the points $(0, r)$ and $(h, 0)$ will be useful.*



Example 6.2.5

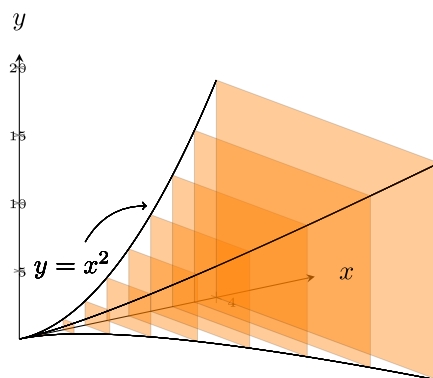
Set up an integral to find the volume of the given solid. The base of the solid is the region bounded by $x = 0$, $x = 4$, and $y = x^2$. The cross-sections are perpendicular to the x -axis and are squares.

Set up an integral to find the volume of the given solid with the same base and square cross-sections, but the cross-sections are perpendicular to the y -axis.



Solution

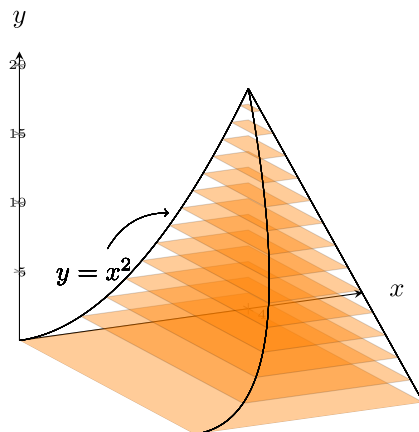
Sketch the region first, then



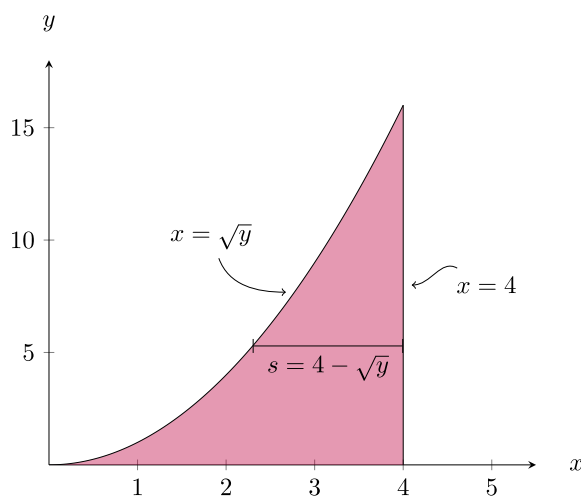
Where we see the side length of the square cross-sections is x^2 , and the slices fill the space from $x = 0$ to $x = 4$ so the definite integral we need to evaluate for the volume is

$$V = \int_0^4 (x^2)^2 dx = \int_0^4 x^4 dx.$$

For the second part of the problem, we are now looking at the region below:



Notice that the slices now go from bottom to top, when the cross-sections are perpendicular to y -axis, we want to use an integral in y . This region has square cross-sections where we can think about “right curve minus left curve” to find the side length of those squares:



Here the square side stretches between $x = \sqrt{y}$ and the vertical line $x = 4$. So, the side length is $s = 4 - \sqrt{y}$. This solid is formed up of slices from $y = 0$ to $y = 16$. Then the integral we solve for the volume is

$$V = \int_0^{16} (4 - \sqrt{y})^2 dy.$$

Exercise 6.2.6

Predict if you think the volumes of the two solids above will be the same or different. Then, evaluate each integral and compare. Were you correct or incorrect in your prediction? If you were incorrect, that is okay, but try to identify why/what went wrong.

Remark

In these problems, the variable of integration is determined based on whether the cross-sections are perpendicular to the x -axis or y -axis.

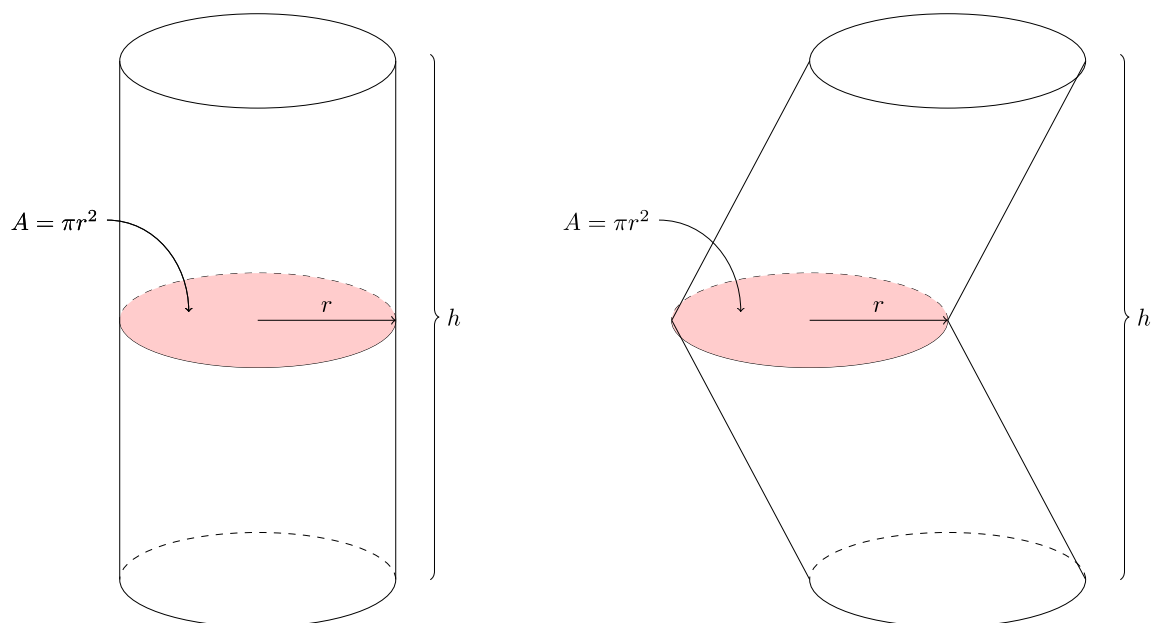
- Slices perpendicular to the x -axis (where slices stack from one x value to another) mean the variable of integration is x .
- Slices perpendicular to the y -axis (where slices stack from one y value to another) mean the variable of integration is y .

Property 6.2.7 (Cavalieri's Principle)

Solids with equal *height* and identical *cross-sectional area* at each height have the same volume.

Remark

These 2 cylinder-like solids have the same volume by Cavalieri's Principle



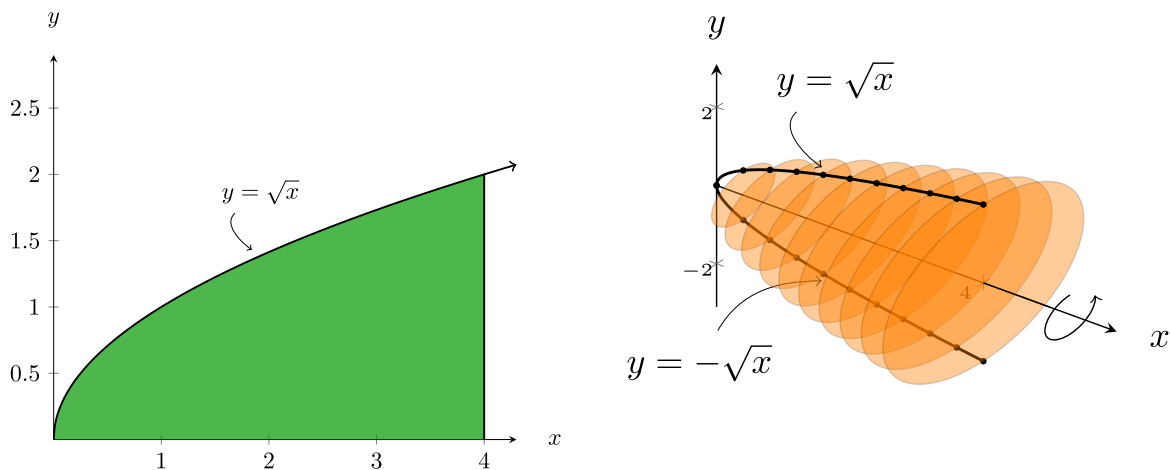
Exercise 6.2.8

Set up an integral to find the volume of the given solid. The base of the solid is the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$. The cross-sections perpendicular to the x -axis are equilateral triangles.

6.2.2. Solids of Revolution: The Disk method**Definition 6.2.9**

A *solid of revolution* is the solid generated by rotating (or revolving) a plane region about an axis in its plane.

Consider the function $y = \sqrt{x}$ on $[0, 4]$. Imagine rotating it all the way around the x -axis to form a solid:

**Note**

If we wish to find the volume of this solid, we have only added one step to our process from before. After that we have the same type of problem as the ones with the cross-sections. We have to build the 2-D region we want to form the solid from.

1. Sketch the function and its reflection about the axis you are told to revolve around. This with the domain give you the 2-D region we go from.
2. Imagine circular cross-sections that are perpendicular to the same axis you just used.
3. Continuing as before the cross-sections will *always* be circular and have radius equal to the function value.

We can make this more specific below.

Theorem 6.2.10 (Volume by Disks for Rotation about the x -axis)

To find the volume of a solid formed by rotating a function $y = f(x)$ on $[a, b]$ about the x -axis first note that the area of each cross-section is $A(x) = \pi(f(x))^2$, then the volume is

$$V = \int_a^b A(x) \, dx = \int_a^b \pi(f(x))^2 \, dx.$$

Exercise 6.2.11

Write a similar statement to the one above for rotating a function $y = f(x)$ for y values in $[a, b]$ about the y -axis. *Hint: careful with the variables you use in the expression. The variable of integration should be the variable used in the integrand.*

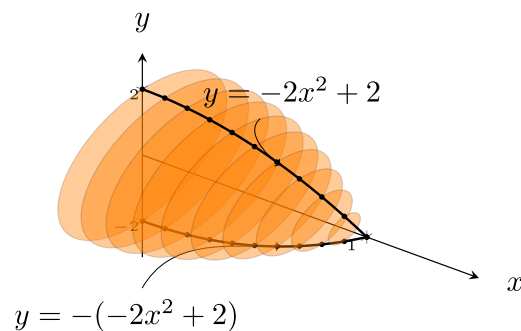
Example 6.2.12

Set up an integral to find the volume of the solid generated by revolving the region in the first quadrant bounded above by $y = -2x^2 + 2$ around the x -axis.

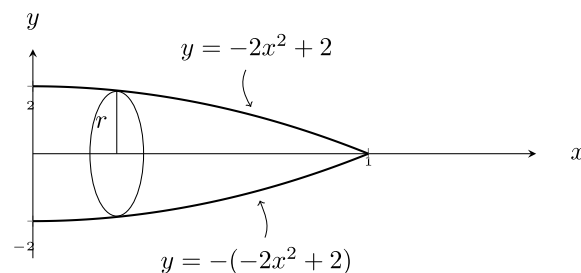
Set up an integral to find the volume when revolving this same region around the y -axis.

 Solution

The solid looks like



However, this might be difficult to draw by hand. An alternative is to just draw the curve in 2-D, its reflection, and a single cross-section to help visualize the radius of the disks. This looks like



Then, we need to find where the curve hits the x -axis. Sometimes we can tell from the graph we draw, sometimes we have to solve for it

$$\begin{aligned} -2x^2 + 2 &= 0 \\ -2(x^2 - 1) &= 0 \\ -2(x + 1)(x - 1) &= 0 \\ x &= 1, -1 \end{aligned}$$

Since we are looking in the first quadrant, $x = 1$ is where this intersection happens (This is one of the bounds for the integral later). The area of the circular cross-section is $\pi r^2 = \pi(-2x^2 + 2)^2$. Thus, the integral we need is $V = \int_0^1 \pi(-2x^2 + 2)^2 dx$.

Exercise 6.2.13

Evaluate the integral above to find the volume of the solid.

Exercise 6.2.14

Set up and solve an integral to find the volume when revolving this same region around the y -axis.

Theorem 6.2.15 (Volume by Disks for Rotation about the y -axis)

To find the volume of a solid formed by rotating a function $y = f(x)$ (and solving for $x = f^{-1}(y)$) for x in $[a, b]$ and y between $f(a)$ and $f(b)$ about the y -axis first note that the area of each cross-section is $A(y) = \pi(f^{-1}(y))^2$, then the volume is

$$V = \int_{f(a)}^{f(b)} A(y) dy = \int_{f(a)}^{f(b)} \pi(f^{-1}(y))^2 dy.$$

In other words,

Note**Finding the Volume of a Solid of Revolution: Disk Method**

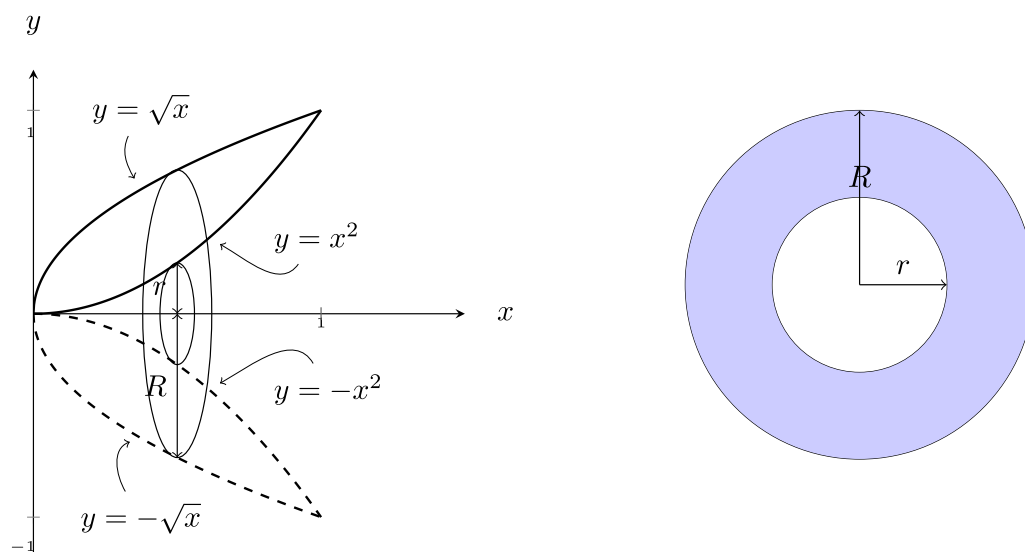
1. Sketch both the region that is being rotated and the axis of revolution.
2. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
3. Determine the limits of integration. They must match the variable of integration!
4. Derive a formula for the radius of the disk.
5. Integrate. Remember units in your final answer when applicable.

6.2.3. Solids of Revolution: The Washer Method

Note that we can only use the disk method if the cross-sections of the solid of revolution are circles. This means that there is no space gap between the region and the axis of revolution. What if there is a gap between the region and the axis of revolution?

Example 6.2.16

Set up an integral to find the volume of the solid generated by revolving the region in the first quadrant bounded by $y = x^2$ and $y = \sqrt{x}$ around the x -axis.

 Solution


We note the curves intersect at $(1, 1)$. There are a few ways to think about this problem, but for now let's go back to thinking about the area of the cross-section and integrating that from 0 to 1 in x . For the cross-sections, the area is a circle with radius $R = \sqrt{x}$ minus the area of a smaller circle with area $r = x^2$. This region is sketched above in blue, and is known as a washer (as in the metal bit used with a nut and bolt that has the same shape). The area of the full region is the difference in areas between the 2 circles. That is, $A(x) = \pi R(x)^2 - \pi r(x)^2$. Then, integrating along the axis perpendicular to the cross-sectional area, we see the volume is

$$\begin{aligned}
 V &= \int_0^1 A(x) \, dx \\
 &= \int_0^1 \pi R(x)^2 - \pi r(x)^2 \, dx \\
 &= \pi \int_0^1 [(\sqrt{x})^2 - (x^2)^2] \, dx \\
 &= \pi \int_0^1 x - x^4 \, dx
 \end{aligned}$$

Exercise 6.2.17

Evaluate the integral above.

Theorem 6.2.18 (Volume by Washers for Rotation about the x -axis)

For a solid of rotation formed by rotating a region around the x -axis where you have the required bounds and radii in terms of x , the volume is

$$V = \int_a^b A(x) \, dx = \pi \int_a^b [(R(x))^2 - (r(x))^2] \, dx$$

Theorem 6.2.19 (Volume by Washers for Rotation about the y -axis)

For a solid of rotation formed by rotating a region around the y -axis where you have the required bounds and radii in terms of y , the volume is

$$V = \int_a^b A(y) \, dy = \pi \int_a^b [(R(y))^2 - (r(y))^2] \, dy$$

In other words,

Note**Finding the Volume of a Solid of Revolution: Washer Method**

1. Sketch both the region that is being rotated and the axis of revolution.
2. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
3. Determine the limits of integration. They must match the variable of integration!
4. Derive a formula for the radii of the washer: $R(x)$ and $r(x)$ (or $R(y)$ and $r(y)$ respectively).
 - These are found by finding the distances of each curve from the axis of revolution. For more, see area between curves section.
5. Integrate. Remember units in your final answer when applicable.

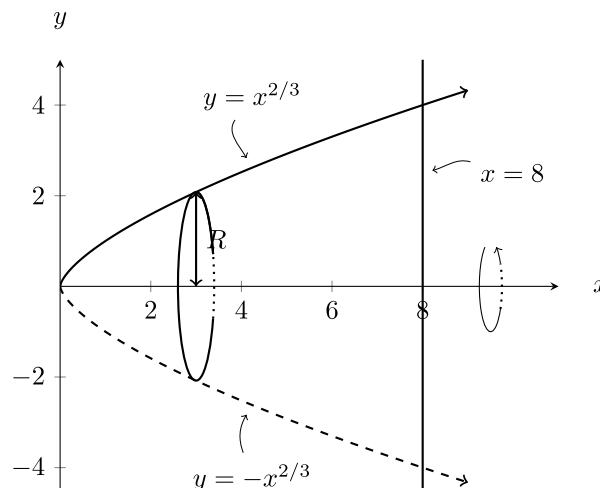
Example 6.2.20

Set up an integral to find the volume of the solid generated by revolving the region bounded by $y = x^{2/3}$, the x -axis, and $x = 8$

- a) around the x -axis
- b) around the y -axis
- c) around the line $y = 6$.

Solution

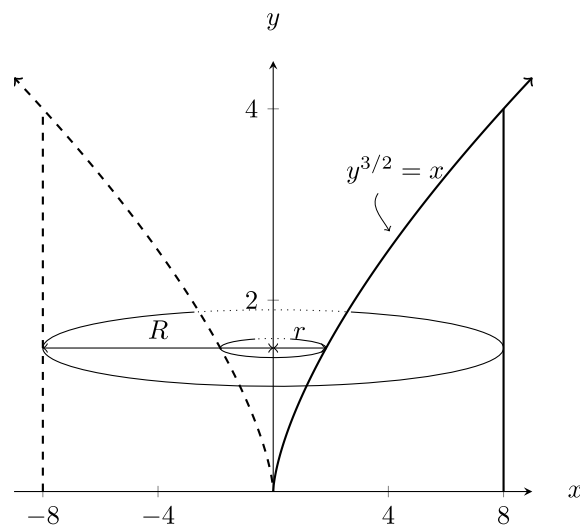
- a) Sketching the region and an idea of the volume we make by rotating around the x -axis we see the cross-sections are disks.



Thus, the volume can be found with the disk method: by thinking of the cross-sectional area as a bunch of circles perpendicular to the x -axis, from $x = 0$ to $x = 8$.

$$\begin{aligned} V &= \pi \int_0^8 (R(x))^2 dx \\ &= \pi \int_0^8 \left(x^{\frac{2}{3}}\right)^2 dx \\ &= \pi \int_0^8 x^{\frac{4}{3}} dx \end{aligned}$$

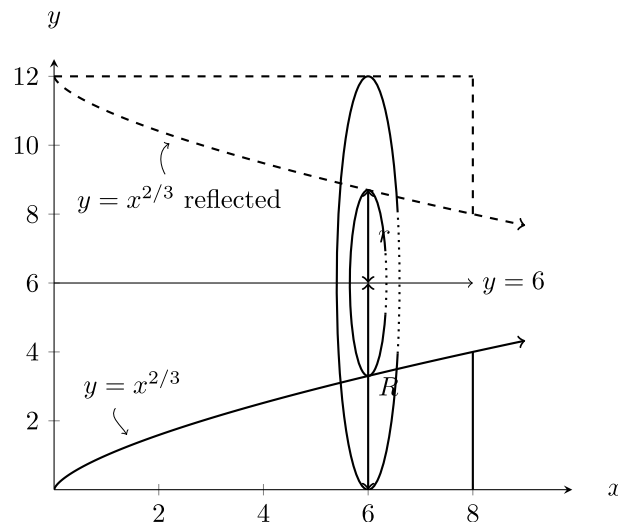
- b) Sketching the region and an idea of the volume we make by rotating around the y -axis we see the cross-sections are washers now.



Notice that we need to express the radii in terms of y now, the large radius is always $R(y) = 8$ and the small radius is $r(y) = y^{\frac{3}{2}}$ which we got from solving the original equation for x . Lastly, we note the solid fills space for y in $[0, 4]$. Then the volume integral will be

$$\begin{aligned} V &= \int_0^4 \pi [(R(y))^2 - (r(y))^2] dy \\ &= \int_0^4 \pi [(8)^2 - (y^{\frac{3}{2}})^2] dy \\ &= \pi \int_0^4 [64 - y^3] dy \end{aligned}$$

- c) Sketching the region and the horizontal line $y = 6$ we have



We first find the radii, or the distance from $y = 6$ to the edges of the region. Note R is always 6, and $r = 6 - x^{\frac{2}{3}}$. Then the area of each washer as a function of x is

$$\begin{aligned} A(x) &= \pi R^2 - \pi r^2 \\ &= \pi \left[6^2 - \left(6 - x^{\frac{2}{3}} \right)^2 \right] \end{aligned}$$

The washers fill the space from $x = 0$ to $x = 8$, so the integral we use is

$$V = \pi \int_0^8 \left[6^2 - \left(6 - x^{\frac{2}{3}} \right)^2 \right] dx$$

6.2 Section Summary:

- We learned how to find the volumes of solids rotated about different horizontal and vertical lines.
- We used the area of a cross-section to develop disk and washer method for finding volumes of the solids of rotation.

6.3. Volumes by Cylindrical Shells

6.3 Learning Objectives:

- I can use shell method to find the volume of solids of rotation.
- I can identify scenarios when shell or washer method is required for a problem based on its geometry.

Exercise 6.3.1

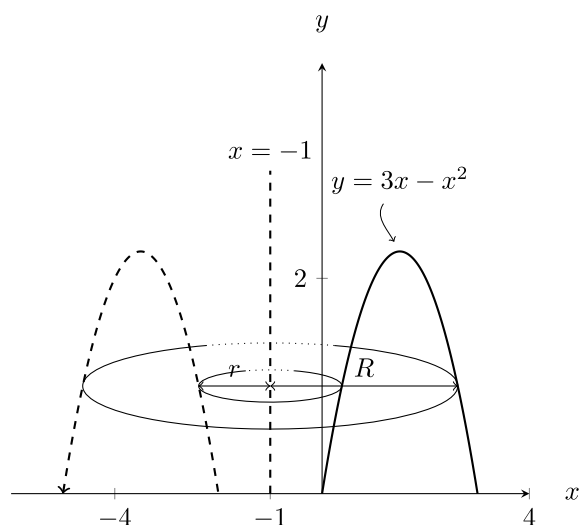
Spend 5-10 minutes trying to set up the integral for using washer method for the following problem. Then stop and analyze why this is particularly difficult.

Rotate the region enclosed by the x -axis and the parabola $y = 3x - x^2$ around the line $x = -1$.



Solution

Note the plot of this region below



helps illustrate the difficulty. Simply put: We have no easy way of determining the radii, r and R .

In the disk and washer method, the cross-sections taken were circular and perpendicular to the axis of rotation. We will introduce a new method, using a new shape for cross-sections, which are taken parallel to the axis of rotation.

Consider the following cookie cutters.

For the next method, we will think of finding the area of a concentric shells making up the shape, then integrating from the center of the shells to the edge to “add” up all the areas that go into the volume.

6.3 Section Summary:

6.4. Work

6.4 Learning Objectives:

6.4 Section Summary: